

Probit Offset with a pace factor

probit family, scored against the incumbent Offset model on the same margins on 2015-2026, 5 seed(s), from unrecorded commit

Contents

1. Model identity and provenance

2. Executive summary

3. Comparator and evaluation population

4. Overall probability performance

5. Year-by-year probability performance

6. Calibration and sharpness

7. Top-choice movement
8. Ensemble stability

9. Feature and correction effects

10. Representative races

11. Betting and bankroll analysis

12. Simulation and uncertainty

13. Technical appendix

14. Limitations and the retrospective-odds warning

Model identity and provenance

Every figure below is read from a declared artifact. This report fits nothing, loads no booster, recomputes no prediction and calls no betting optimiser.

MODEL Probit Offset with a pace factor probit	COMPARATOR the incumbent Offset model on the same margins	SEEDS 5 7, 17, 42, 123, 256	TEST YEARS 12 2015-2026
BOOSTING ROUNDS n/a	SOURCE COMMIT n/a clean		

Config digest unrecorded, model view bagged_probit_offset_pace, artifacts read from /Users/jlcheung/PycharmProjects/hkjc-mod/results/probit_models.

ENVIRONMENT

Component	Version
numpy	2.5.2
pandas	3.0.5
python	3.12.11
xgboost	3.4.1

Deltas in this report use model_minus_market_log_loss: race_multinomial_log_loss, where **negative** is better. That is stated because three delta fields existed in this repository and they did not all point the same way.

Executive summary

MEAN DELTA -0.000785 negative is better	YEARS BETTER 10/11	WORST YEAR 0.001175	CALIBRATED DELTA -0.000341 both sides calibrated
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Probit Offset with a pace factor is better than its comparator by 0.000785 race_multinomial_log_loss, measured against the incumbent Offset model on the same margins on the evaluated races. The direction holds in 10 of 11 evaluated years. Every seed agrees on the sign of the improvement.

Comparator and evaluation population

Same comparator as the market-anchored probit, so the two are directly comparable to each other: the only difference between them is the shared pace factor in the covariance.

TEST RACES
8,675

YEARS
12
2015-2026

SEEDS
5

HONEST-SCALE YEARS
11
years whose utility scale
was fitted rather than
fallen back

Some years fell back rather than fitting their own scale

2015 did not produce an honest utility scale (insufficient_history). Those years are excluded from the headline delta and from the bankroll paths, and shown separately in the year-by-year table: a fallback scale is not a measurement of the candidate.

Overall probability performance

races	model log loss	comparator log loss	market log loss	honest delta	better year share
8675	2.024605	2.025390	2.028805	-0.000785	0.909091

One row, pooled over every evaluated race. The per-race distribution is what the year-by-year section shows; a pooled mean can be carried by a handful of races and this figure alone cannot tell you whether it was.

Year-by-year probability performance

test year	races	model log loss	comparator log loss	market log loss	delta	honest
2015	742	2.015068	1.992494	1.998720	0.022574	no
2016	744	2.092451	2.093824	2.099277	-0.001373	yes
2017	772	2.017101	2.018056	2.021559	-0.000955	yes
2018	756	2.057580	2.057714	2.061504	-0.000134	yes
2019	759	1.927395	1.926220	1.932827	0.001175	yes
2020	799	2.006427	2.007792	2.010400	-0.001365	yes
2021	783	2.001404	2.001934	2.006659	-0.000530	yes
2022	783	2.050094	2.051758	2.051195	-0.001663	yes
2023	796	2.020665	2.021139	2.024066	-0.000475	yes
2024	792	2.017508	2.017590	2.025744	-0.000083	yes
2025	748	2.051190	2.052113	2.057397	-0.000923	yes
2026	201	2.028838	2.031148	2.026224	-0.002310	yes



Ahead of its comparator in 10 of 12 years. 1 row(s) are italicised as partial or fallback years, shown rather than dropped so the evaluation population stays visible.

Calibration and sharpness

This architecture reports calibration through a paired comparison rather than binned reliability: candidate and comparator are each given a temperature on the same races, so the calibrated delta is the like-for-like figure. Fallback-scale years are excluded here, as they are from the headline.

test year	candidate calibrated loss	comparator calibrated loss	calibrated delta
2016	2.092447	2.093236	-0.000789
2017	2.018158	2.018225	-0.000067
2018	2.058258	2.057516	0.000742
2019	1.929684	1.927971	0.001712
2020	2.007505	2.009214	-0.001709
2021	2.000953	2.001587	-0.000633
2022	2.050874	2.051204	-0.000330
2023	2.021438	2.021954	-0.000517
2024	2.017259	2.017686	-0.000427
2025	2.053116	2.053894	-0.000779

test year	candidate calibrated loss	comparator calibrated loss	calibrated delta
2026	2.029934	2.030893	-0.000959

If the calibrated delta is materially smaller than the uncalibrated one, part of the candidate's advantage was a temperature the comparator could also have had.

Top-choice movement

Not applicable to this model. Top-choice movement compares a model's first pick against the market's. This candidate is scored against its own comparator on fixed mean utilities, not against the market, so a market top-choice comparison would answer a different question than the rest of this report.

Ensemble stability

Not applicable to this model. There is no ensemble to be stable. This architecture is a choice *link* applied to one set of mean utilities, not a bag of independently fitted members, so seed variation here is integration noise rather than model disagreement -- and that is reported in the appendix instead.

Feature and correction effects

Not applicable to this model. There are no per-feature effects to attribute. The link changes how utilities become probabilities; the utilities themselves come from the comparator's fit, whose feature effects are reported in that model's own report.

Representative races

Not applicable to this model. Representative races are chosen by how much a model's correction moved a runner's probability. This candidate applies a link to fixed mean utilities, so the movement to explain belongs to the comparator's fit and is shown in that model's report. A race chosen here would be chosen by integration noise.

Betting and bankroll analysis

Retrospective odds

Every stake, return and bankroll figure in this report is computed against historical result-page odds and official dividends. Those are not an executable pre-race price: they are the prices after the pool closed, and a bet placed at them could not have been placed. Nothing here establishes profitability. A profitability claim requires a locked model shadow-tested against timestamped pre-race odds snapshots.

Do not select a threshold from these returns.

Every year through 2026 has been used as development data in this repository: features were searched, architectures compared and thresholds inspected against these same seasons. The chronological split is real, so a test year is genuinely after its training rows, but no test year is unseen. Treat the forward figures as probability progress rather than as a locked out-of-sample result, and do not select a Kelly level or an edge threshold from this reused history. The confirmation step that this run cannot substitute for is a prospective shadow test against archived pre-race price snapshots.

SETTLEMENT BY STAKING RULE

strategy	n bets	turnover	net return	roi	total log growth	max drawdown	final bankroll	underwater days
flat_stake@edge>=0.02	2478	24,780,000	2,437,000	0.098345	0.218091	0.141115	12,437,000	1591
kelly_no_rebate@0.01	3074	5,425,050	460189.000000	0.084827	0.044991	0.032880	10,460,189	2020
kelly_no_rebate@0.02	3098	11,032,740	901968.500000	0.081754	0.086358	0.064842	10901968.500000	2041
kelly_no_rebate@0.05	3105	28,777,360	2096664.500000	0.072858	0.190345	0.155180	12096664.500000	2048
kelly_no_rebate@0.1	3109	60,224,300	3542814.000000	0.058827	0.303271	0.288615	13,542,814	2052
kelly_settled_rebate@0.01	3075	5,469,950	464131.000000	0.084851	0.059712	0.030272	10,615,306	2021
kelly_settled_rebate@0.02	3100	11,379,020	936,365	0.082289	0.141064	0.054746	11,514,988	2039
kelly_settled_rebate@0.05	3105	31,616,240	2354086.500000	0.074458	0.357380	0.127510	14295795.500000	2042
kelly_settled_rebate@0.1	3112	73,559,100	4528827.500000	0.061567	0.663634	0.239878	19418356.500000	2049
rebate_aware_kelly@0.01	3401	8,565,860	303853.500000	0.035473	0.075352	0.038809	10782631.600000	2020
rebate_aware_kelly@0.02	3522	20,174,080	660,765	0.032753	0.179440	0.078998	11965470.250000	2102
rebate_aware_kelly@0.05	4659	70,260,090	1169248.500000	0.016642	0.484687	0.188781	16236669.350000	3018
rebate_aware_kelly@0.1	6053	182,638,970	-764049.000000	-0.004183	0.825703	0.340403	22834849.550000	4301

Model view bagged_probit_offset_pace. Seed -1. Every rule is settled on the same selected bets at the same odds, so the differences between these rows are stake sizing alone.

What this settlement can decide

Secondary diagnostic. This suite selects on paired race log loss; betting returns cannot select a model and are not evidence for or against a probit candidate.

EXPECTED AGAINST REALISED, BY ESTIMATED EDGE

FLAT STAKE

bucket	mean estimated edge	expected return	realised at edge>=0.02	bets at edge>=0.02
0.020000 to 0.050000	0.033504	0.033504	-0.0190	644
0.050000 to 0.100000	0.073145	0.073145	-0.1448	543
0.100000 to 0.200000	0.143710	0.143710	0.0307	527
0.200000 to unbounded	0.512179	0.512179	0.4167	764

KELLY, NO REBATE

bucket	mean estimated edge	expected return	realised at 0.01	realised at 0.02	realised at 0.05	realised at 0.1	bets at 0.01	bets at 0.02	bets at 0.05	bets at 0.1
0 to 0.020000	0.009974	0.009974	-0.1079	-0.1053	-0.0953	-0.0835	596	620	627	631
0.020000 to 0.050000	0.033504	0.033504	-0.0778	-0.0776	-0.0817	-0.0843	644	644	644	644
0.050000 to 0.100000	0.073145	0.073145	0.0600	0.0603	0.0595	0.0567	543	543	543	543
0.100000 to 0.200000	0.143710	0.143710	-0.1321	-0.1319	-0.1325	-0.1332	527	527	527	527
0.200000 to unbounded	0.512179	0.512179	0.2229	0.2173	0.2024	0.1780	764	764	764	764

KELLY, SETTLED REBATE

bucket	mean estimated edge	expected return	realised at 0.01	realised at 0.02	realised at 0.05	realised at 0.1	bets at 0.01	bets at 0.02	bets at 0.05	bets at 0.1
0 to 0.020000	0.009959	0.009959	-0.1085	-0.1022	-0.0864	-0.0661	597	622	627	634
0.020000 to 0.050000	0.033504	0.033504	-0.0784	-0.0810	-0.0868	-0.0941	644	644	644	644
0.050000 to 0.100000	0.073145	0.073145	0.0615	0.0616	0.0629	0.0628	543	543	543	543
0.100000 to 0.200000	0.143710	0.143710	-0.1324	-0.1325	-0.1326	-0.1323	527	527	527	527
0.200000 to unbounded	0.512179	0.512179	0.2229	0.2190	0.2063	0.1849	764	764	764	764

REBATE-AWARE KELLY

bucket	mean estimated edge	expected return	realised at 0.01	realised at 0.02	realised at 0.05	realised at 0.1	bets at 0.01	bets at 0.02	bets at 0.05	bets at 0.1
unbounded to 0	-0.032449	-0.032449	-0.0686	-0.0790	-0.0013	-0.0240	325	422	1554	2950
0 to 0.020000	0.009954	0.009954	-0.1785	0.0267	-0.0462	-0.0484	598	622	627	625
0.020000 to 0.050000	0.033504	0.033504	-0.0080	-0.1223	-0.1232	-0.1485	644	644	644	644
0.050000 to 0.100000	0.073145	0.073145	0.0047	0.0308	-0.0100	0.0109	543	543	543	543
0.100000 to 0.200000	0.143710	0.143710	-0.1355	-0.1223	-0.1229	-0.1310	527	527	527	527
0.200000 to unbounded	0.512179	0.512179	0.1820	0.1861	0.1882	0.1565	764	764	764	764

The check that matters most in this section: if the realised return in each edge bucket does not track the expected one, the edge estimate is wrong and every stake sized from it is wrong with it. The bucket counts are small enough that one winner moves a bucket, so read the pattern across buckets rather than any single number.

A negative-edge bucket appears only under rebate-aware sizing, and only at the larger fractions. That is the rule working as defined rather than a filter failing: the rebate is a cliff, so a runner whose own edge is negative can pay for itself by pushing the race's turnover over the threshold, and the stakes only get large enough for that at the higher fractions.

KELLY PRESETS

OPENING BANKROLL

HK\$10,000,000

PRESETS SETTLED

1% Kelly, 2% Kelly, 5% Kelly, 10% Kelly

PATH MODES

annual reset, continuous

SEEDS

5
7, 17, 42, 123, 256

YEARS STAKED

2015-2026

Each path is rebased on its own opening balance, so zero means break-even for every line from the same point. Where a year is excluded, a continuous path opens at the balance the bank held when the kept years begin and an annual-reset path opens at the declared bankroll, so the two show different opening figures for the same rule. Rebate policy: 10% of the race net loss, once that loss reaches HK\$10,000, as a cliff. Flat staking is omitted from these panels. Its stake is a fixed amount per bet and this settlement never halts on ruin, so its path can pass below zero -- a measure of how badly a rule loses rather than a balance an account could hold. It is also not a Kelly fraction, so it belongs to no preset. Its settlement is in the strategy table above. 2015 is excluded from every path below. These test years have no earlier year to fit the utility scale on and fall back to the predeclared theoretical value. The utility is divided by the scale, so the fallback amplifies corrections where a fitted scale halves them: their stakes and claimed edges are several times any fitted year's and are not comparable with them. Exclude these years from any return figure -- filter on `is_honest_scale` in the per-run CSV.

1% KELLY

OPENING BANKROLL

HK\$10,000,000

BEST FINAL RETURN

+0.069%
Rebate-aware Kelly, annual reset

BETS

1,438
settled bets

REBATE RECEIVED

HK\$12,003
the most any rule received

■ Kelly, no rebate ■ Kelly, settled rebate ■ Rebate-aware Kelly



Cumulative bankroll return at 1% Kelly Stepped at each settlement: a bankroll moves only when a bet settles, so the path is flat between settlements and a smooth line would draw growth on days nothing was staked. Dashed lines are the annual-reset mode. Retrospective odds.

▼ Table view

Series	Settlements	Final return	Note
Kelly, no rebate, annual reset	1,325	+0.064%	1,427 bets
Kelly, no rebate, continuous	1,325	-0.258%	1,427 bets
Kelly, settled rebate, annual reset	1,326	+0.065%	1,428 bets
Kelly, settled rebate, continuous	1,326	-0.258%	1,428 bets
Rebate-aware Kelly, annual reset	1,327	+0.069%	1,438 bets
Rebate-aware Kelly, continuous	1,327	-0.361%	1,438 bets

BETS PER YEAR

Year	Kelly, no rebate	Kelly, settled rebate	Rebate-aware Kelly
2016	212	212	215
2017	213	213	213
2018	174	174	177
2019	116	116	116
2020	193	193	196
2021	149	150	150
2022	141	141	142
2023	48	48	48
2024	81	81	81
2025	80	80	80
2026	20	20	20

2026 is a season still in progress, italicised: a shorter year, not a decline in activity. Kelly, no rebate: 212 bets in 2016 against 80 in 2025. Kelly, settled rebate: 212 bets in 2016 against 80 in 2025. Rebate-aware Kelly: 215 bets in 2016 against 80 in 2025. Where the path is flat and these counts are low, the model stopped finding bets rather than losing an edge it had.

Rebate received: Rebate-aware Kelly, annual reset HK\$12,003; Rebate-aware Kelly, continuous HK\$12,003. A rule that reaches the threshold gets money back on a losing race, which makes some races worth betting that otherwise are not

-- which is why its bet count can differ from the others'. Both path modes are shown: continuous carries the bankroll across years, while annual-reset restarts it each year, so the reset mode measures per-year skill without compounding.

2% KELLY

OPENING BANKROLL

HK\$10,000,000

BEST FINAL RETURN

+0.268%

Rebate-aware Kelly,
annual reset

BETS

1,561

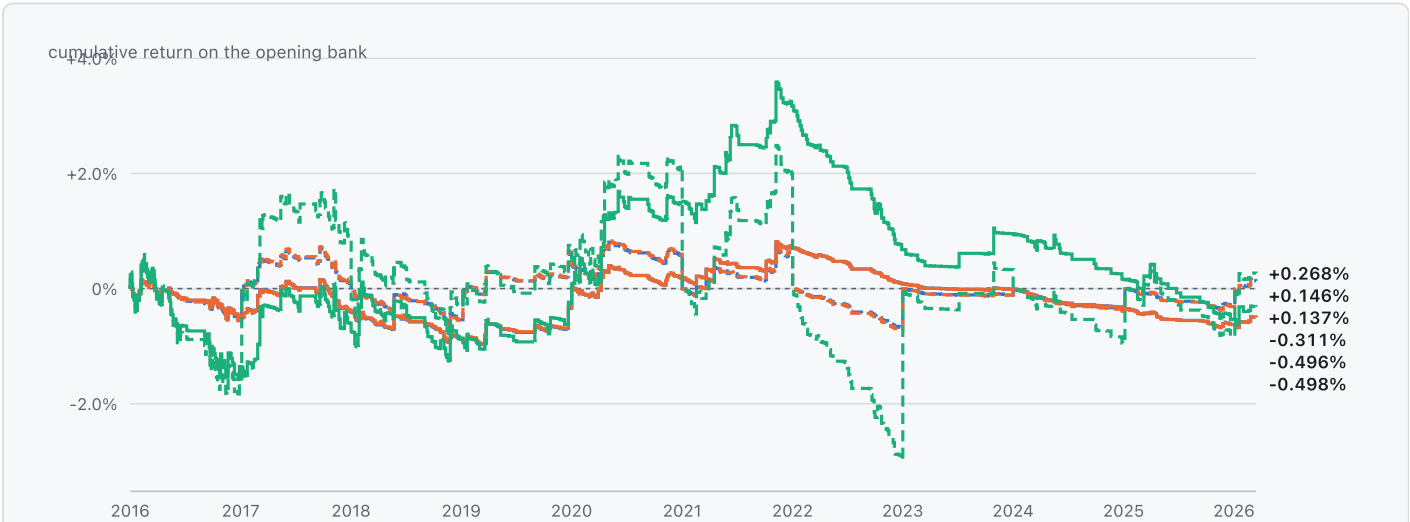
settled bets

REBATE RECEIVED

HK\$242,009

the most any rule
received

■ Kelly, no rebate ■ Kelly, settled rebate ■ Rebate-aware Kelly



Cumulative bankroll return at 2% Kelly Stepped at each settlement: a bankroll moves only when a bet settles, so the path is flat between settlements and a smooth line would draw growth on days nothing was staked. Dashed lines are the annual-reset mode. Retrospective odds.

▼ Table view

Series	Settlements	Final return	Note
Kelly, no rebate, annual reset	1,346	+0.137%	1,451 bets
Kelly, no rebate, continuous	1,346	-0.496%	1,451 bets
Kelly, settled rebate, annual reset	1,348	+0.146%	1,453 bets
Kelly, settled rebate, continuous	1,348	-0.498%	1,453 bets
Rebate-aware Kelly, annual reset	1,418	+0.268%	1,561 bets
Rebate-aware Kelly, continuous	1,418	-0.311%	1,561 bets

BETS PER YEAR

Year	Kelly, no rebate	Kelly, settled rebate	Rebate-aware Kelly
2016	214	215	245
2017	215	215	225
2018	175	175	190
2019	121	121	126
2020	198	198	210
2021	153	153	158

Year	Kelly, no rebate	Kelly, settled rebate	Rebate-aware Kelly
2022	144	145	159
2023	48	48	51
2024	83	83	91
2025	80	80	84
2026	20	20	22

2026 is a season still in progress, italicised: a shorter year, not a decline in activity. Kelly, no rebate: 214 bets in 2016 against 80 in 2025. Kelly, settled rebate: 215 bets in 2016 against 80 in 2025. Rebate-aware Kelly: 245 bets in 2016 against 84 in 2025. Where the path is flat and these counts are low, the model stopped finding bets rather than losing an edge it had.

Rebate received: Rebate-aware Kelly, annual reset HK\$242,009; Rebate-aware Kelly, continuous HK\$242,009; Kelly, settled rebate, annual reset HK\$2,092; Kelly, settled rebate, continuous HK\$2,092. A rule that reaches the threshold gets money back on a losing race, which makes some races worth betting that otherwise are not -- which is why its bet count can differ from the others'. Both path modes are shown: continuous carries the bankroll across years, while annual-reset restarts it each year, so the reset mode measures per-year skill without compounding.

5% KELLY

OPENING BANKROLL

HK\$10,000,000

BEST FINAL RETURN

+8.3%

Rebate-aware Kelly,
continuous

BETS

2,693

settled bets

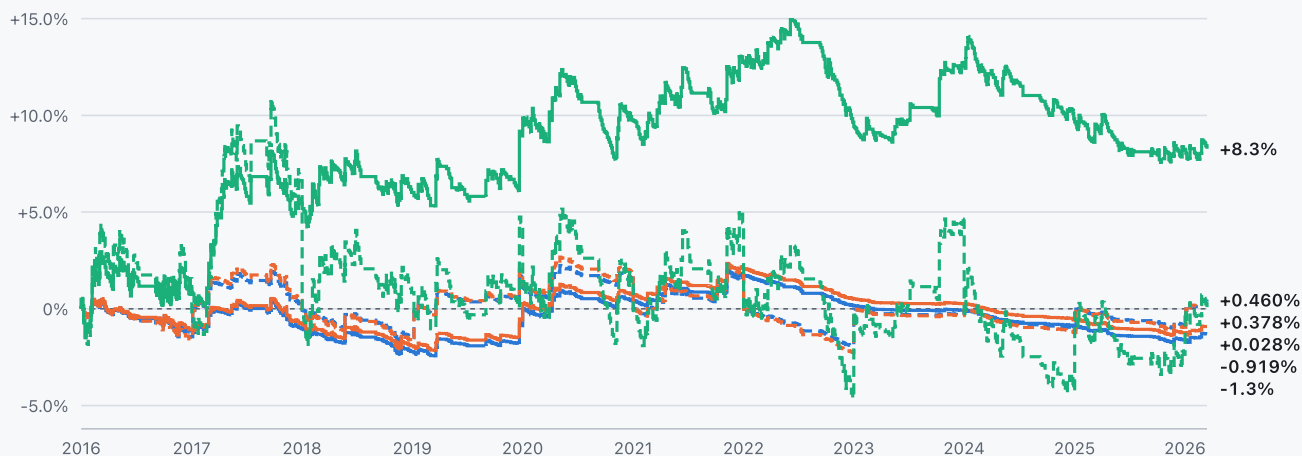
REBATE RECEIVED

HK\$2,116,768

the most any rule
received

Kelly, no rebate Kelly, settled rebate Rebate-aware Kelly

cumulative return on the opening bank



Cumulative bankroll return at 5% Kelly Stepped at each settlement: a bankroll moves only when a bet settles, so the path is flat between settlements and a smooth line would draw growth on days nothing was staked. Dashed lines are the annual-reset mode. Retrospective odds.

▼ Table view

Series	Settlements	Final return	Note
Kelly, no rebate, annual reset	1,353	+0.378%	1,458 bets
Kelly, no rebate, continuous	1,353	-1.3%	1,458 bets
Kelly, settled rebate, annual reset	1,353	+0.460%	1,458 bets
Kelly, settled rebate, continuous	1,353	-0.919%	1,458 bets
Rebate-aware Kelly, annual reset	2,344	+0.028%	2,693 bets
Rebate-aware Kelly, continuous	2,344	+8.3%	2,693 bets

BETS PER YEAR

Year	Kelly, no rebate	Kelly, settled rebate	Rebate-aware Kelly
2016	216	216	366
2017	217	217	332
2018	175	175	299
2019	121	121	193
2020	199	199	327
2021	154	154	296
2022	145	145	292
2023	48	48	143
2024	83	83	210
2025	80	80	190
2026	20	20	45

2026 is a season still in progress, italicised: a shorter year, not a decline in activity. Kelly, no rebate and Kelly, settled rebate: 216 bets in 2016 against 80 in 2025. Rebate-aware Kelly: 366 bets in 2016 against 190 in 2025. Where the path is flat and these counts are low, the model stopped finding bets rather than losing an edge it had.

Rebate received: Rebate-aware Kelly, annual reset HK\$2,116,768; Rebate-aware Kelly, continuous HK\$2,116,768; Kelly, settled rebate, annual reset HK\$54,867; Kelly, settled rebate, continuous HK\$54,867. A rule that reaches the threshold gets money back on a losing race, which makes some races worth betting that otherwise are not -- which is why its

bet count can differ from the others'. Both path modes are shown: continuous carries the bankroll across years, while annual-reset restarts it each year, so the reset mode measures per-year skill without compounding.

10% KELLY

OPENING BANKROLL

HK\$10,000,000

BEST FINAL RETURN

+10.4%

Rebate-aware Kelly,
continuous

BETS

4,079

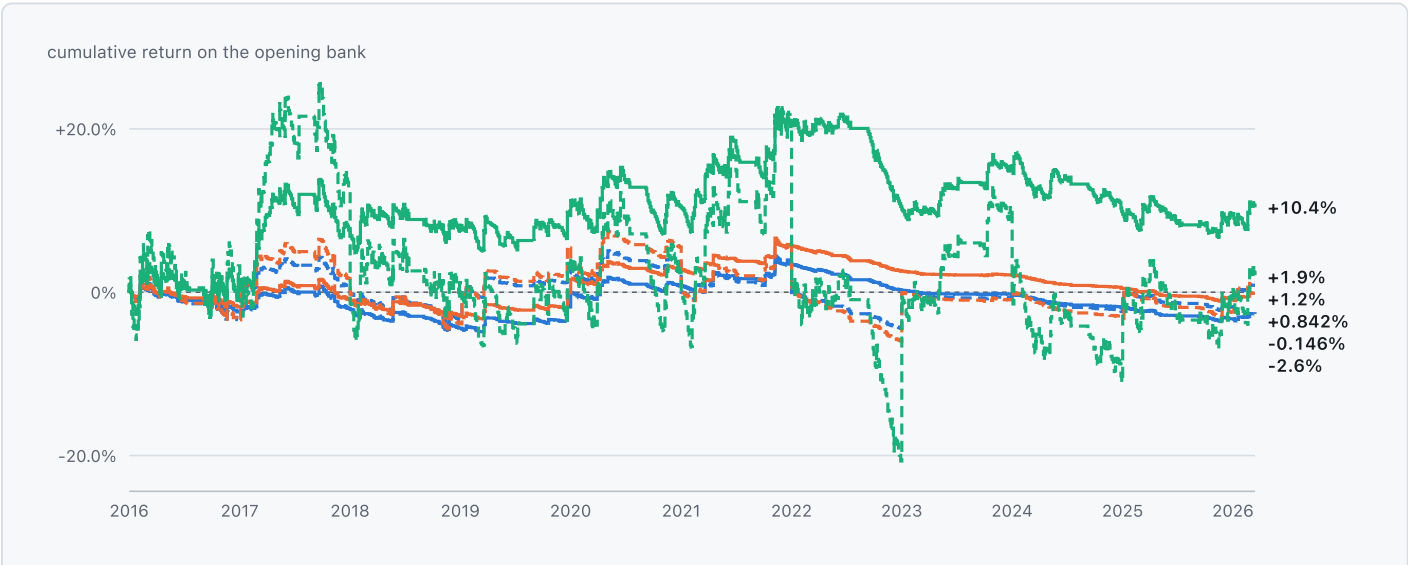
settled bets

REBATE RECEIVED

HK\$6,819,004

the most any rule
received

Kelly, no rebate Kelly, settled rebate Rebate-aware Kelly



Cumulative bankroll return at 10% Kelly Stepped at each settlement: a bankroll moves only when a bet settles, so the path is flat between settlements and a smooth line would draw growth on days nothing was staked. Dashed lines are the annual-reset mode. Retrospective odds.

Table view

Series	Settlements	Final return	Note
Kelly, no rebate, annual reset	1,357	+0.842%	1,462 bets
Kelly, no rebate, continuous	1,357	-2.6%	1,462 bets
Kelly, settled rebate, annual reset	1,359	+1.2%	1,465 bets
Kelly, settled rebate, continuous	1,359	-0.146%	1,465 bets
Rebate-aware Kelly, annual reset	3,612	+1.9%	4,079 bets
Rebate-aware Kelly, continuous	3,612	+10.4%	4,079 bets

BETS PER YEAR

Year	Kelly, no rebate	Kelly, settled rebate	Rebate-aware Kelly
2016	218	218	487
2017	217	217	456
2018	175	176	399
2019	121	121	316
2020	199	199	470
2021	154	154	433

Year	Kelly, no rebate	Kelly, settled rebate	Rebate-aware Kelly
2022	145	146	433
2023	49	49	279
2024	83	83	359
2025	80	80	350
2026	21	22	97

2026 is a season still in progress, italicised: a shorter year, not a decline in activity. Kelly, no rebate: 218 bets in 2016 against 80 in 2025. Kelly, settled rebate: 218 bets in 2016 against 80 in 2025. Rebate-aware Kelly: 487 bets in 2016 against 350 in 2025. Where the path is flat and these counts are low, the model stopped finding bets rather than losing an edge it had.

Rebate received: Rebate-aware Kelly, annual reset HK\$6,819,004; Rebate-aware Kelly, continuous HK\$6,819,004; Kelly, settled rebate, annual reset HK\$497,019; Kelly, settled rebate, continuous HK\$497,019. A rule that reaches the threshold gets money back on a losing race, which makes some races worth betting that otherwise are not -- which is why its bet count can differ from the others'. Both path modes are shown: continuous carries the bankroll across years, while annual-reset restarts it each year, so the reset mode measures per-year skill without compounding.

PRESET SENSITIVITY

Preset	Rule and mode	Opening	Final	Return	Bets	Rebate
1% Kelly	Kelly, no rebate, annual reset	HK\$10,000,000	HK\$10,006,394	+0.064%	1,427	HK\$0
1% Kelly	Kelly, no rebate, continuous	HK\$10,487,232	HK\$10,460,189	-0.258%	1,427	HK\$0
1% Kelly	Kelly, settled rebate, annual reset	HK\$10,000,000	HK\$10,006,512	+0.065%	1,428	HK\$0
1% Kelly	Kelly, settled rebate, continuous	HK\$10,642,758	HK\$10,615,306	-0.258%	1,428	HK\$0
1% Kelly	Rebate-aware Kelly, annual reset	HK\$10,000,000	HK\$10,006,890	+0.069%	1,438	HK\$12,003
1% Kelly	Rebate-aware Kelly, continuous	HK\$10,821,686	HK\$10,782,632	-0.361%	1,438	HK\$12,003
2% Kelly	Kelly, no rebate, annual reset	HK\$10,000,000	HK\$10,013,666	+0.137%	1,451	HK\$0
2% Kelly	Kelly, no rebate, continuous	HK\$10,956,335	HK\$10,901,968	-0.496%	1,451	HK\$0
2% Kelly	Kelly, settled rebate, annual reset	HK\$10,000,000	HK\$10,014,600	+0.146%	1,453	HK\$2,092
2% Kelly	Kelly, settled rebate, continuous	HK\$11,572,618	HK\$11,514,988	-0.498%	1,453	HK\$2,092
2% Kelly	Rebate-aware Kelly, annual reset	HK\$10,000,000	HK\$10,026,794	+0.268%	1,561	HK\$242,009
2% Kelly	Rebate-aware Kelly, continuous	HK\$12,002,829	HK\$11,965,470	-0.311%	1,561	HK\$242,009
5% Kelly	Kelly, no rebate, annual reset	HK\$10,000,000	HK\$10,037,836	+0.378%	1,458	HK\$0
5% Kelly	Kelly, no rebate, continuous	HK\$12,254,918	HK\$12,096,664	-1.3%	1,458	HK\$0
5% Kelly	Kelly, settled rebate, annual reset	HK\$10,000,000	HK\$10,046,033	+0.460%	1,458	HK\$54,867
5% Kelly	Kelly, settled rebate, continuous	HK\$14,428,359	HK\$14,295,796	-0.919%	1,458	HK\$54,867
5% Kelly	Rebate-aware Kelly, annual reset	HK\$10,000,000	HK\$10,002,850	+0.028%	2,693	HK\$2,116,768
5% Kelly	Rebate-aware Kelly, continuous	HK\$14,995,525	HK\$16,236,669	+8.3%	2,693	HK\$2,116,768
10% Kelly	Kelly, no rebate, annual reset	HK\$10,000,000	HK\$10,084,190	+0.842%	1,462	HK\$0
10% Kelly	Kelly, no rebate, continuous	HK\$13,911,384	HK\$13,542,814	-2.6%	1,462	HK\$0
10% Kelly	Kelly, settled rebate, annual reset	HK\$10,000,000	HK\$10,124,954	+1.2%	1,465	HK\$497,019
10% Kelly	Kelly, settled rebate, continuous	HK\$19,446,768	HK\$19,418,356	-0.146%	1,465	HK\$497,019
10% Kelly	Rebate-aware Kelly, annual reset	HK\$10,000,000	HK\$10,188,580	+1.9%	4,079	HK\$6,819,004
10% Kelly	Rebate-aware Kelly, continuous	HK\$20,683,884	HK\$22,834,850	+10.4%	4,079	HK\$6,819,004

The spread across presets is the point: a result that appears at only one Kelly fraction is a result about that fraction rather than about the model. A larger fraction stakes more of the bank on the same edge estimate, so it amplifies both the edge and the error in it.

Simulation and uncertainty

MEAN DELTA
-0.000677

INTERVAL
-0.0012 to
-0.0002
80% level

EXCLUDES ZERO
FAVOURABLY
yes

RACE DAYS
891
39,665 races

REPLICATES
2,000
seed 20260824

A day-block bootstrap: race days are resampled within each test year, so the interval accounts for clustering within a meeting, and the interval lies entirely below zero. It conditions on the fitted model and does *not* include refit uncertainty.

What a simulation cannot tell you

The model-assumed simulation draws winners from the model's own probabilities. If those probabilities are wrong the simulation is confidently wrong in the same direction, so it bounds sampling risk and says nothing about model risk.

Technical appendix

THE UTILITY SCALE

FITTED MINIMUM
1.9932

FITTED MAXIMUM
2.2280

THEORY
0.7797
the value theory predicts

The fitted scale sits above the value theory predicts, in every year: 1.993 to 2.228 against 0.780 (the scale the fallback year used, which is the predeclared theoretical value). Theory gets this wrong and the fit is what the report uses -- a scale asserted from theory would divide the utilities by too little, and at that value the market-relative candidate loses instead of winning. A run that took the theoretical value on faith would have reported the wrong sign, which is why the first year -- which has no history and falls back to it -- is excluded from the headline and labelled.

CONFIGURATION

Item	Value
Architecture	probit_offset
Comparator field	offset
Integration	exact_gauss_hermite_quadrature
Shared factors	1
Factor loadings	['early_style_ewm']

Item	Value
Opening bankroll	10,000,000
Bankroll mode	continuous
Rebate-aware sizing	no

PROMOTION GATES

Gate	Verdict
1. paired delta is negative	pass
2. interval excludes zero favourably	pass
3. no year degrades beyond tolerance	FAIL
4. survives nested calibration	pass
5. direction stable across seeds	pass
6. improvement exceeds integration error 10x	pass

5 of 6 checkable gate assertions hold. Passing them is not promotion. An interaction diagnostic across predeclared cohorts, reproduction from a clean identified commit, and confirmation on a future prospective period are not derivable from these artifacts. Production remains the five-seed Offset ensemble until they are met.

PER-SEED DELTA

Seed	Delta
7	-0.000809
17	-0.000759
42	-0.000705
123	-0.000778
256	-0.000875

5 of 5 seeds favour the candidate, and every seed agrees on the sign. Shown individually rather than only pooled, because a mean whose seeds disagree is a different claim from one whose seeds agree.

UTILITY SCALE

setting	value	note
2015	0.779697	fitted on strictly earlier years; fallback to the theoretical value
2016	1.993223	fitted on strictly earlier years
2017	2.034700	fitted on strictly earlier years
2018	2.210128	fitted on strictly earlier years

FACTOR SCALE GRID

setting	value	metric	note
2016_scale_0.00	0	1.992242	race log loss on the fitting window; zero means no factor
2016_scale_0.15	0.150000	1.992222	race log loss on the fitting window; zero means no factor
2016_scale_0.30	0.300000	1.992200	race log loss on the fitting window; zero means no factor
2016_scale_0.50	0.500000	1.992205	race log loss on the fitting window; zero means no factor
2016_scale_0.80	0.800000	1.992289	race log loss on the fitting window; zero means no factor
2017_scale_0.00	0	2.042575	race log loss on the fitting window; zero means no factor
2017_scale_0.15	0.150000	2.042517	race log loss on the fitting window; zero means no factor

setting	value	metric	note
2017_scale_0.30	0.300000	2.042405	race log loss on the fitting window; zero means no factor
2017_scale_0.50	0.500000	2.042286	race log loss on the fitting window; zero means no factor
2017_scale_0.80	0.800000	2.042203	race log loss on the fitting window; zero means no factor
2018_scale_0.00	0	2.034032	race log loss on the fitting window; zero means no factor
2018_scale_0.15	0.150000	2.033987	race log loss on the fitting window; zero means no factor
2018_scale_0.30	0.300000	2.033898	race log loss on the fitting window; zero means no factor
2018_scale_0.50	0.500000	2.033805	race log loss on the fitting window; zero means no factor
2018_scale_0.80	0.800000	2.033756	race log loss on the fitting window; zero means no factor
2019_scale_0.00	0	2.055180	race log loss on the fitting window; zero means no factor
2019_scale_0.15	0.150000	2.055151	race log loss on the fitting window; zero means no factor
2019_scale_0.30	0.300000	2.055109	race log loss on the fitting window; zero means no factor
2019_scale_0.50	0.500000	2.055108	race log loss on the fitting window; zero means no factor
2019_scale_0.80	0.800000	2.055190	race log loss on the fitting window; zero means no factor

24 row(s) from the sensitivity artifact, in 2 study/studies. Where a study varies an integration setting, a result that moves with it is a property of the integration rather than of the model.

Limitations and the retrospective-odds warning

Retrospective odds, restated

Every stake, return and bankroll figure in this report is computed against historical result-page odds and official dividends. Those are not an executable pre-race price: they are the prices after the pool closed, and a bet placed at them could not have been placed. Nothing here establishes profitability. A profitability claim requires a locked model shadow-tested against timestamped pre-race odds snapshots.

Restated at the end as well as beside the betting numbers, deliberately. It is the single most load-bearing caveat in this report and the one a reader skimming for a return figure is most likely to miss.

Every test year is also development data

Every year through 2026 has been used as development data in this repository: features were searched, architectures compared and thresholds inspected against these same seasons. The chronological split is real, so a test year is genuinely after its training rows, but no test year is unseen. Treat the forward figures as probability progress rather than as a locked out-of-sample result, and do not select a Kelly level or an edge threshold from this reused history. The confirmation step that this run cannot substitute for is a prospective shadow test against archived pre-race price snapshots.

WHAT ELSE THIS REPORT DOES NOT ESTABLISH

- It measures Probit Offset with a pace factor against the incumbent Offset model on the same margins. Beating a comparator and beating the market are different claims.
- Its intervals condition on the fitted model, so they bound sampling risk and not model risk.
- Every figure is out of sample in time but in sample in *design*: the feature set, the hyperparameters and the staking rules were all chosen with these years visible.
- Nothing here is a shadow test. A profitability claim needs a locked model run forward against timestamped pre-race prices.